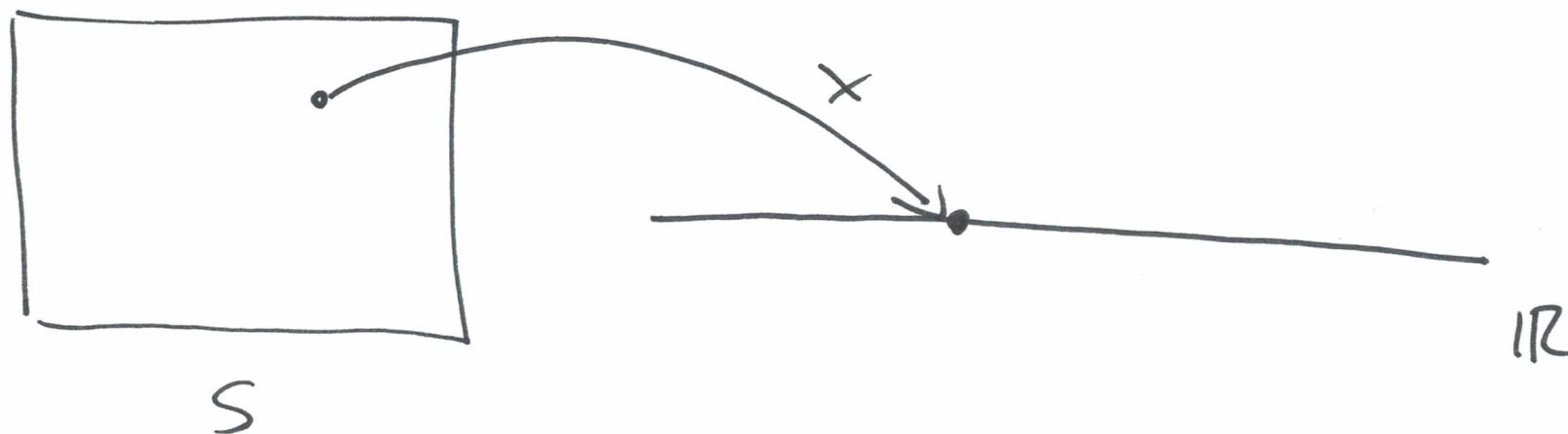


DISCRETE RANDOM VARIABLES:

A RANDOM VARIABLE X MAPS AN
OUTCOME FROM THE SAMPLE SPACE TO
A REAL NUMBER.



KEY CONCEPT: PROBABILITY MASS
FUNCTION (PMF)

$$P_X(x) = P(X=x)$$

Diagram illustrating the notation for the Probability Mass Function (PMF):

- P : LITTLE P (indicated by an upward arrow)
- X : BIG X (indicated by an upward arrow)
- (x) : LITTLE x (indicated by an upward arrow)
- P : BIG P (indicated by an upward arrow)
- X : BIG X (indicated by a downward arrow)
- (x) : LITTLE x (indicated by an upward arrow)

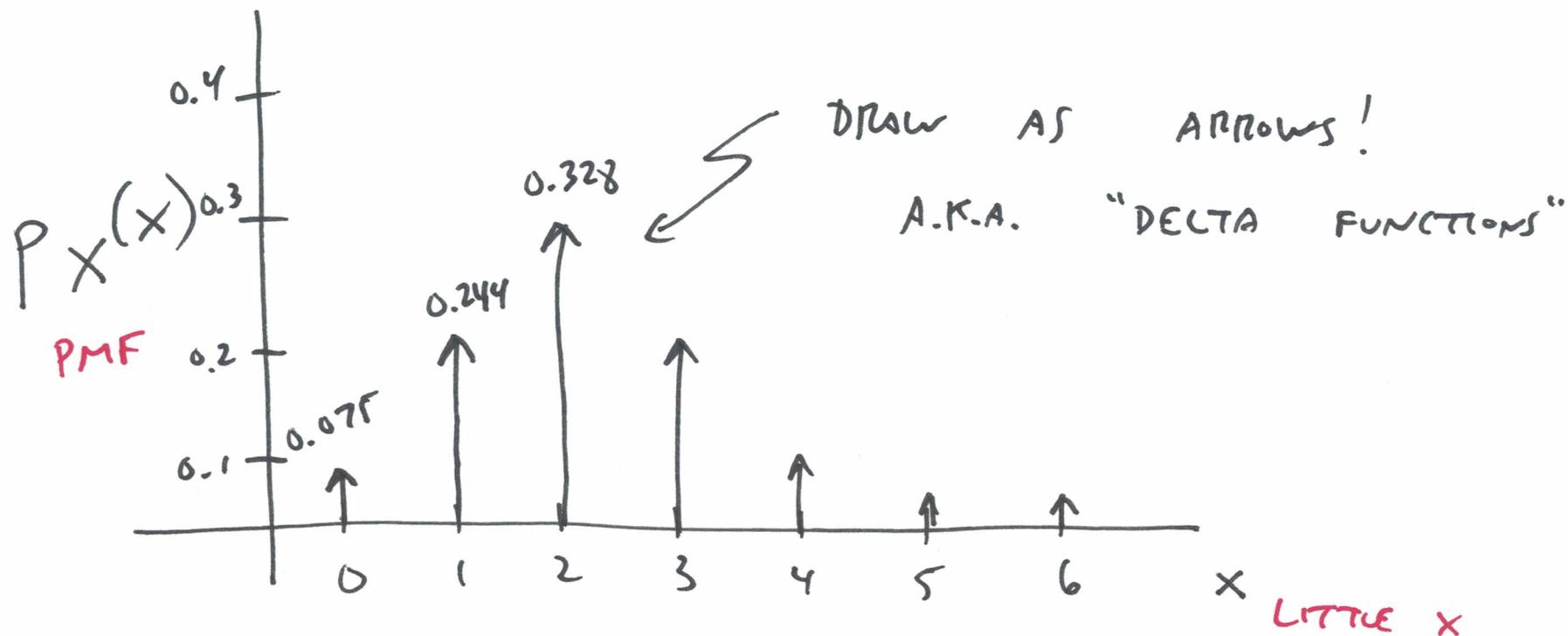
$$P_Y(2) = P(Y=2)$$

EX 6 CARS. $P(\text{CAT ON CAR}) = 0.35$

$X = \#$ OF CATS ON CARS.

X HAS SAMPLE SPACE $\{0, 1, 2, 3, 4, 5, 6\}$

DRAW PMF OF X :



$$P_X(0) = P(X=0) = (0.65)^6 = 0.075$$

$$P_X(1) = P(X=1) = \binom{6}{1} (0.35) (0.65)^5 = 0.244$$

$$P_X(2) = P(X=2) = \binom{6}{2} (0.35)^2 (0.65)^4 = 0.328$$

⋮

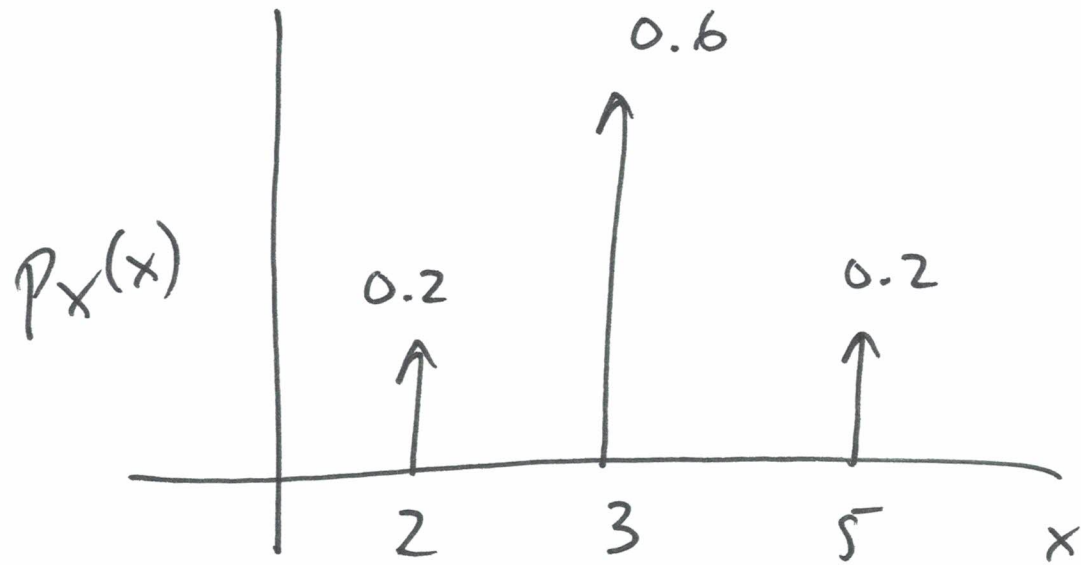
0.236

0.095

0.021

0.002

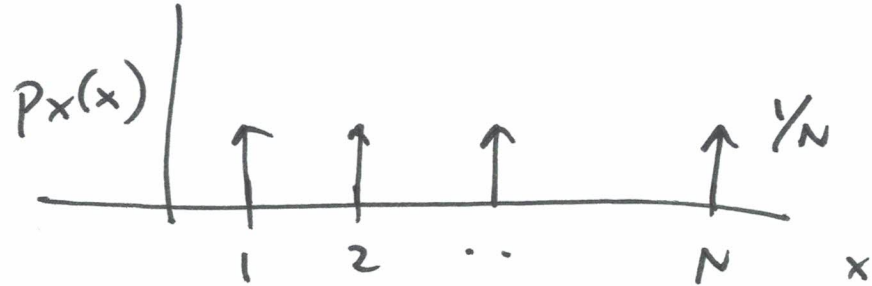
~~Exercise~~



Common PMFs:

UNIFORM RANDOM VARIABLE ON $\{1, 2, \dots, N\}$

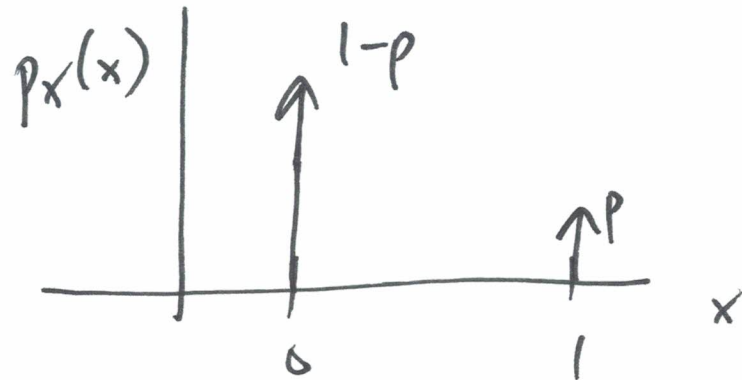
$$P_X(x) = \frac{1}{N}$$



BERNOULLI RANDOM VARIABLE (RV)

FAIL $P_X(0) = 1 - p$

SUCCESS $P_X(1) = p$



BINOMIAL

RV: # OF SUCCESSES IN n
BERNOULLI TRIALS.

$$P_X(k) = P(k \text{ SUCCESSES IN } n \text{ TRIALS})$$

$$= \binom{n}{k} (p)^k (1-p)^{n-k}$$

$$k \in \{0, 1, \dots, n\}$$

GEOMETRIC

RV : # OF TRIALS UNTIL 1ST SUCCESS

$$P_X(k) = (1-p)^{k-1} p \quad k \in \{1, 2, \dots, \infty\}$$

$$P_X(1) = p$$

$$P_X(2) = (1-p)p \quad \text{ETC..}$$

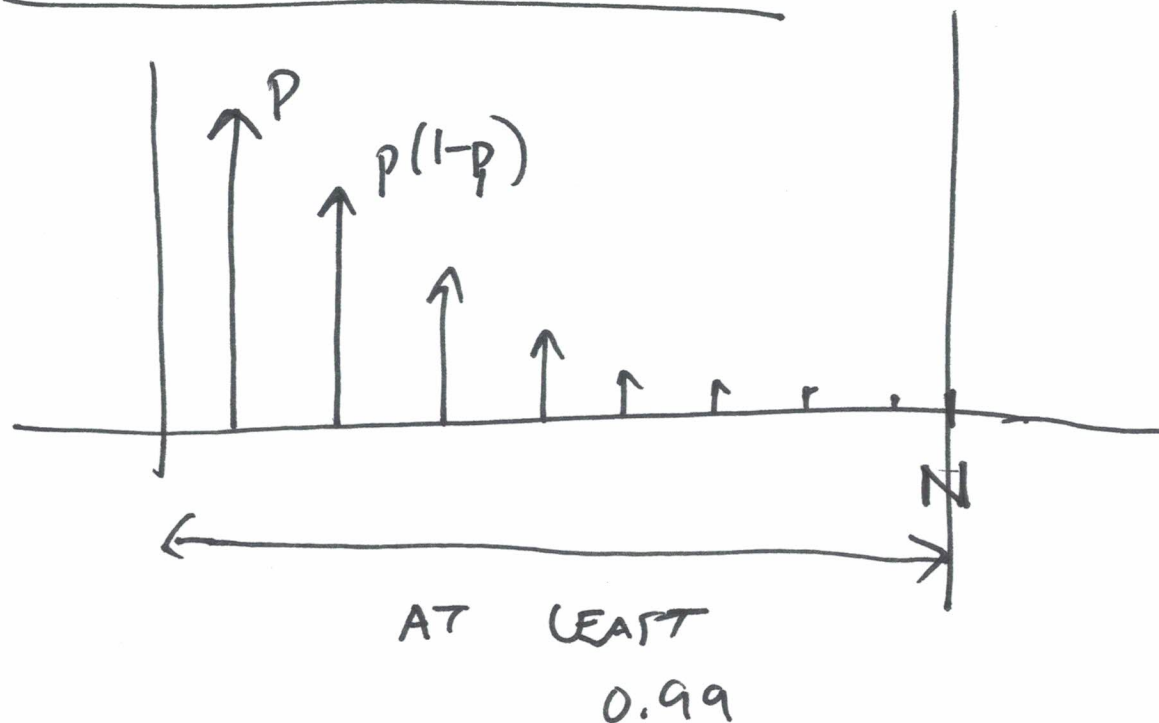
WALKWAY BLOCKED

$$P(\text{FIX ON A GIVEN DAY}) = 0.02$$

$X = \#$ DAYS UNTIL WALKWAY IS OPEN.

FIND N SO THAT

$$P(X > N) < 0.01$$



$$P_X(n) = (0.98)^{n-1} (0.02)$$

$$\sum_{n=N+1}^{\infty} p_X(n) < 0.01$$

$$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}$$

$$= \sum_{n=N+1}^{\infty} (0.98)^{n-1} (0.02)$$

$$= (0.02) (0.98)^N \underbrace{\sum_{n=0}^{\infty} (0.98)^n}$$

$$= (0.02) (0.98)^N \frac{1}{1-0.98}$$

$$= (0.98)^N < 0.01$$

$$N \log 0.98 < \log 0.01$$

$$N > \frac{\log 0.01}{\log 0.98}$$

$$= \underline{\underline{227.9}}$$

$$\sum_{n=0}^{\infty} a^n =$$

$$\sum_{n=0}^{\infty} a^n - \sum_{n=N+1}^{\infty} a^n$$

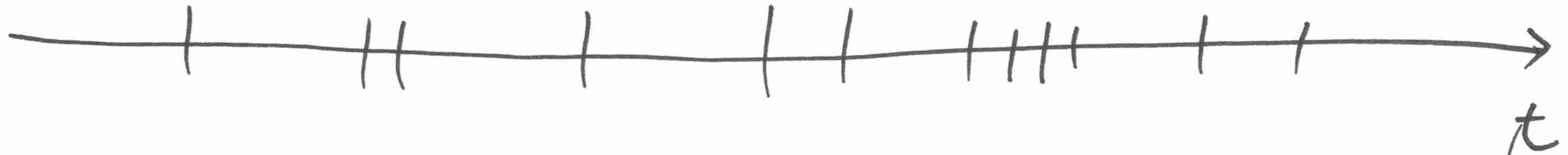
$$\sum_{n=0}^{\infty} a^n - \sum_{n=N+1}^{\infty} a^n$$

$$= \frac{1}{1-a} - a^{N+1} \sum_{n=0}^{\infty} a^n$$

$$= \frac{1 - a^{N+1}}{1 - a}$$

POISSON RANDOM VARIABLE

USED TO MODEL ARRIVALS / EVENTS IN
A GIVEN TIME PERIOD



$$P_X(k) = \frac{\alpha^k}{k!} e^{-\alpha}$$

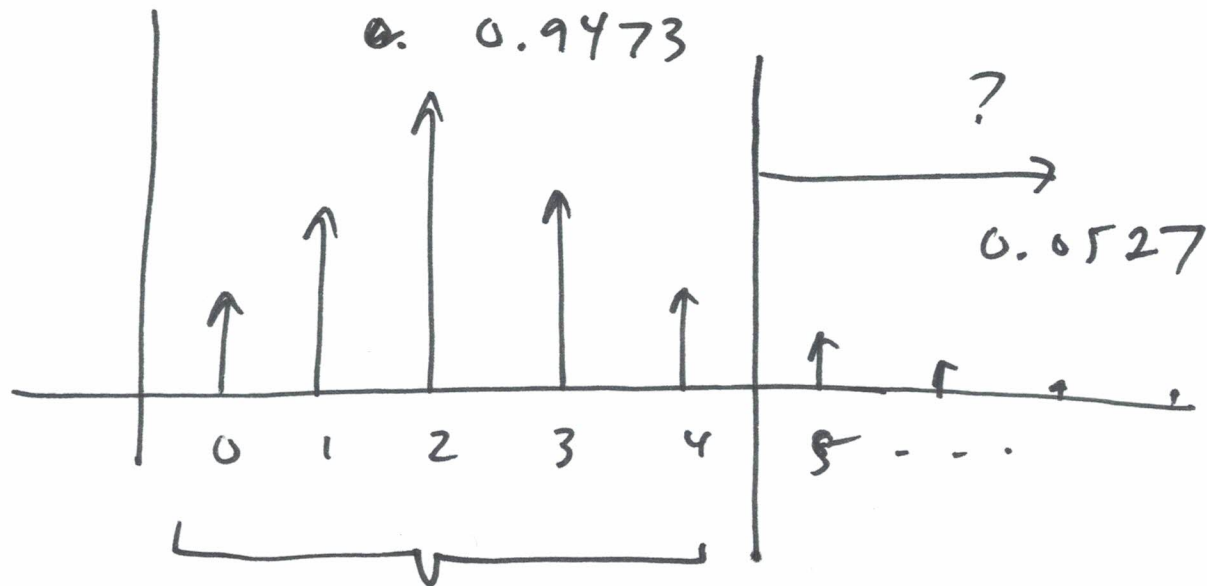
$$k = 0, 1, 2, \dots$$

α = AVERAGE NUMBER OF EVENTS IN DESIRED
PERIOD OF TIME.

e.g. POST ON RPI REDDIT AVERAGE
1 POST / 4 HOURS.

$P(\text{MORE THAN 4 POSTS IN 8 HOURS?})$

$\alpha = 2$ (EXPECT AVG. OF 2 POSTS IN 8 HOURS).



$$P_X(k) = \frac{2^k}{k!} e^{-2}$$

$$P_X(0) = \frac{2^0}{0!} e^{-2} = e^{-2}$$

$$P_X(1) = \frac{2^1}{1!} e^{-2} = +2e^{-2}$$

$$P_X(2) = \frac{2^2}{2!} + 2e^{-2}$$

$$P_X(3) = \frac{2^3}{3!} + \frac{8}{6} e^{-2}$$

$$P_X(4) = \frac{2^4}{4!} + \frac{16}{24} e^{-2}$$

$$A = 7e^{-2}$$

$$1 - 7e^{-2} = 0.0527$$

EXAM 1

THURSDAY FEB 12

IN CLASS 2:00 - 3:20 PM

ONE ONE-SIDED HANDWRITTEN PAGE OF NOTES

SIMPLE CALCULATOR

NO PHONES/LAPTOPS

2 BACK EXAMS ~~W~~ ON PIAZZA.